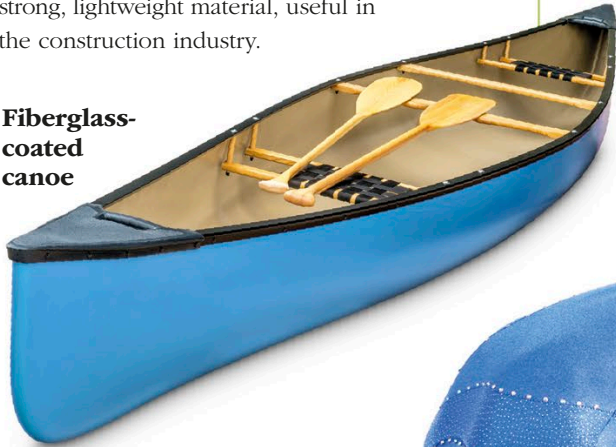


Glass-based insulator

- **What?** Fiberglass
- **Who?** Games Slayter
- **Where and when?** US, 1932

At the Owens-Illinois glass company, Games Slayter discovered a way of mass-producing glass wool, known today as fiberglass. The material traps air, making it ideal for insulation. In 1936, glass wool was combined with a plastic resin to create a strong, lightweight material, useful in the construction industry.

Fiberglass-coated canoe



Elastic sportswear

- **What?** Spandex
- **Who?** Joseph Shivers
- **Where and when?** US, 1958

While working at the DuPont company, the American chemist Joseph Shivers was looking for a light, synthetic material to use in women's clothing. In the 1950s, he found it—an elastic-like fiber that was named spandex, an anagram of “expands.” It was patented in 1958 and marketed as Lycra.



Elastic attire made from spandex is ideal for flexible gymnasts.

Kevlar vests such as this have been used by the US military since the 1980s



Kevlar bulletproof vests are very strong and light.

Tough, protective plastic

- **What?** Kevlar
- **Who?** Stephanie Kwolek and Paul Morgan
- **Where and when?** US, 1965

Kevlar is a plastic that is five times stronger than steel. It was developed by chemists working for DuPont in the US. It is related to another man-made material called nylon, but with an extra chemical to add strength and stiffness. It was first used in racing tires, and is also used in golf clubs and flame-proof clothing.

Flexible electronics

- **What?** Flexible electronic screens
- **Who?** Plastic Logic
- **Where and when?** Germany, 2004

German scientists have discovered a way to make light, thin, and flexible screens that can carry digital information. At present, this technology is used in signage, wristwatches, and other wearable devices, but we might soon have flexible computer screens (left).



FAST FACTS

- Man-made materials are typically much more durable than their natural counterparts. Once an advantage, this is now seen as an environmental problem, as they do not degrade.
- Chinese material scientists have created the world's lightest material called graphene aerogel, which is almost entirely made out of air.